

Name: _____

Date: _____

NUMBER OPERATIONS, (CHECKING ANSWERS USING INVERSE OPERATIONS)

LO: I can use inverse operations to check calculations and verify my answers.

What are inverse operations?

Inverse operations are operations that **undo** each other. Addition undoes subtraction. Multiplication undoes division.

Checking: After solving a calculation, use the inverse to check. If $45 + 32 = 77$, then $77 - 32$ should = 45.

Inverse pairs

●	+ and - are inverses	undo each other
●	× and ÷ are inverses	undo each other
●	If $7 \times 8 = 56$, then $56 \div 8 = 7$	check
●	+ and × are inverses	wrong pairing
●	Inverse of squaring is halving	it is square-rooting

Key idea

Use the inverse operation on your answer to get back to the starting number. If you do, your answer is correct.

$$45 + 32 = 77$$

check: $77 - 32 = 45$

Inverse undoes the operation

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – checking addition

Check: $156 + 89 = 245$

Use inverse: $245 - 89$

$$245 - 89 = 156$$

156 matches correct

Correct

Subtract one addend from the sum; you should get the other.

Example 2 – checking multiplication

Check: $23 \times 15 = 345$

Use inverse: $345 \div 15$

$$345 \div 15 = 23$$

23 matches correct

Correct

Divide the product by one factor; you should get the other.

Example 3 – finding an error

Check: $64 \div 8 = 9$

Use inverse: $9 \times 8 = 72$

$$72 \neq 64$$

The answer should be 8, not 9

Incorrect

The inverse check reveals the error: $64 \div 8 = 8$.

Example 4 – checking subtraction

Check: $500 - 237 = 263$

Use inverse: $263 + 237$

$$263 + 237 = 500$$

500 matches correct

Correct

Add the result to the subtracted number; you should get the original.

YOUR TURN – PRACTICE (deliberate practice)

1. Check additions

Use inverse (subtraction) to check. Write correct or incorrect.

- a) $78 + 45 = 123$
- b) $156 + 89 = 245$
- c) $234 + 178 = 402$
- d) $67 + 88 = 155$

2. Check subtractions

Use inverse (addition) to check.

- a) $200 - 85 = 115$
- b) $463 - 198 = 265$
- c) $500 - 237 = 273$
- d) $1000 - 456 = 544$

3. Check multiplications

Use inverse (division) to check.

- a) $12 \times 15 = 180$
- b) $25 \times 13 = 325$
- c) $34 \times 7 = 248$
- d) $45 \times 20 = 900$

4. Check divisions

Use inverse (multiplication) to check.

- a) $144 \div 12 = 12$
- b) $256 \div 8 = 34$
- c) $630 \div 9 = 70$
- d) $484 \div 22 = 22$

5. Mixed checking

Check each calculation using the inverse. State correct or give the right answer.

a) $56 + 78 = 134$

b) $15 \times 16 = 250$

c) $900 \div 25 = 36$

d) $1000 - 382 = 628$

e) $17 \times 19 = 323$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) The inverse of addition is subtraction ☐

Correct answer: _____

Reason: _____

b) The inverse of multiplication is addition ☐

Correct answer: _____

Reason: _____

c) If $7 \times 8 = 56$, then $56 \div 7 = 8$ ☐

Correct answer: _____

Reason: _____

d) The inverse of squaring is dividing by 2 ☐

Correct answer: _____

Reason: _____

e) Checking answers helps find mistakes ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A student calculates $345 \times 27 = 9315$. Without redoing the full multiplication, use **inverse operations** and **estimation** to check whether this is correct.

Expression: _____

Simplified: _____

b) The equation $x^2 = 49$ has two solutions. Find both. Then use the inverse (squaring) to verify each solution. What is the inverse of squaring?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

